

Maharashtra Public Service Commission Assessment

Question Paper Name :	Deputy Director of Steam Boilers 17th August 2025 Shift 2
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Deputy Director of Steam Boilers

Group Number :	1
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Deputy Director of Steam Boilers

Section Id :	56230331
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	100

Number of Questions to be attempted :	100
Section Marks :	200
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	56230343
Question Shuffling Allowed :	Yes

Question Number : 1 Question Id : 5623032961 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

Which of the following is not true about the latent heat of formation ?

Options :

1. It is the amount of heat required to convert water at given temperature and pressure into steam at same temperature and pressure
2. It is expressed by symbol h_g in the steam table
3. The value of latent heat varies with pressure
4. Latent heat is lower at higher pressure

Question Number : 2 Question Id : 5623032962 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

For steam, _____.

Options :

1. the critical temperature is 221.2°C and critical pressure is 374.15 bar.
2. the critical temperature is 374.15°C and critical pressure is 221.2 bar.
3. the critical temperature is 221.2°C and critical pressure is 221.2 bar.
4. the critical temperature is 374.15°C and critical pressure is 374.15° bar.

Question Number : 3 Question Id : 5623032963 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which of the following is not the process that takes place during Rankine cycle on which steam power plant works ?

Options :

1. Reversible adiabatic expansion in the turbine (or steam engine).
2. Constant pressure transfer of heat in the condenser.
3. Reversible adiabatic pumping process in the feed pump.
4. Constant volume transfer of heat in the boiler.

Question Number : 4 Question Id : 5623032964 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Temporary hardness of water can be removed by boiling the water and it is caused by _____ which decompose when heated into insoluble carbonates. Permanent hardness is caused by salts of _____, which are not affected by boiling.

Options :

1. Calcium sulphate, calcium carbonates
2. Caustic soda, carbon dioxide
3. Calcium carbonates, calcium sulphate
4. All of the above

Question Number : 5 Question Id : 5623032965 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Air from a condenser is extracted from coldest zone because _____.

Options :

1. the amount of air to be handled by the air pump will be low.
2. the air removed from the coldest zone will contain less water vapour.

3. less work is required to operate the air pump.
4. air pump of lower quality can be used.

Question Number : 6 Question Id : 5623032966 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

The actual vacuum in a condenser is equal to _____.

Options :

1. barometric pressure + actual pressure
2. barometric pressure – actual pressure
3. guage pressure + atmospheric pressure
4. guage pressure – atmospheric pressure

Question Number : 7 Question Id : 5623032967 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

Which of the following is the most economical method of producing “make-up water” for high pressure boilers ?

Options :

1. Demineralization

2. Hydrogen zeolite process
3. Sodium zeolite process
4. Deaeration process

Question Number : 8 Question Id : 5623032968 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

_____ softeners are used to remove calcium and magnesium ions that would otherwise form scale in the boilers.

Options :

1. Natural salts
2. Oils
3. Zeolites
4. None of the above

Question Number : 9 Question Id : 5623032969 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

Low pH of boiler water promotes _____, and high pH can lead to both _____ and scaling.

Options :

1. damage, expansion of tubes
2. faster boiling of water, formation of steam
3. corrosion, caustic embrittlement cracking
4. caustic embrittlement cracking, corrosion

Question Number : 10 Question Id : 5623032970 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

In the ideal vapour power cycle, what is the hypothetical or ideal process for the steam turbine ?

Options :

1. Reversible constant pressure heating of water to form a steam.
2. Reversible adiabatic expansion of steam.
3. Reversible constant pressure heat rejection as the steam condenser.
4. Reversible adiabatic compression of liquid.

Question Number : 11 Question Id : 5623032971 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

A body that absorbs all the impinging radiant heat is called as a _____.

Options :

1. white body
2. grey body
3. black body
4. opaque body

Question Number : 12 Question Id : 5623032972 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Entropy is often described as a measure of _____.

Options :

1. Disorder at macroscopic level.
2. Order at microscopic level.
3. Energy transformation efficiency.
4. Temperature variation within a system.

Question Number : 13 Question Id : 5623032973 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The evaporative capacity of a boiler may be expressed in terms of _____.

Options :

1. Kg of steam/h
2. Kg of steam/h/m² of heating surface
3. Kg of steam/kg of fuel fired
4. All of the above

Question Number : 14 Question Id : 5623032974 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The specific heat of a substance at higher pressures or lower specific volumes can be determined from the zero pressure specific heats and the PVT behaviour of the substance by applying _____.

Options :

1. The ideal gas flow.
2. The test of exactness.
3. The clausius-Clapeyron equation.

4. The Maxwell equations.

Question Number : 15 Question Id : 5623032975 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What is one of the potential consequences of thermal stresses in a deaerator ?

Options :

1. Increased steam production
2. Water hammer
3. Reduced dissolved oxygen
4. Enhanced heat transfer

Question Number : 16 Question Id : 5623032976 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

During sub-stoichiometric combustion of a hydrocarbon fuel, _____.

Options :

1. Hydrogen is burnt out in preference to carbon.
2. Carbon content is burnt in preference to hydrogen.

3. Both are burnt at the same rate and same time.

- First carbon is converted into CO which then oxidizes the fuel hydrogen
4. to water.

Question Number : 17 Question Id : 5623032977 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

For Gas Burning, which of the following is false statement ?

Options :

1. Ease of control of furnace temperature
2. Ability to produce a long slow burning flame
3. Lowers level of greenhouse gases than other fossil fuels
4. Difficulty of temperature regulation

Question Number : 18 Question Id : 5623032978 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

When the fuel composition is known, then the air-fuel ratio can be calculated by _____ method.

Options :

1. Carbon balance
2. Hydrogen balance
3. Combined carbon-hydrogen balance
4. All of the above

Question Number : 19 Question Id : 5623032979 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

Which of the following is true about biomass firing in the boilers ?

Options :

1. The content of oxygen in bio-mass is very high compared to coal.
2. Bio-mass pellets may have very low content of sulphur and nitrogen compared to coal which makes them environmental friendly.
3. Ash produced by bio-mass combustion are low in alkali and alkaline earth elements.
4. Volatile matter in bio-mass enhances the fire explosion risk in the preprocessing system.

Question Number : 20 Question Id : 5623032980 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Performance of coal pulverizer used in the boiler depends on which of the following factors ?

Options :

1. Grindability of the coal
2. Surface moisture on the coal
3. Fineness of the grind needed
4. All of the above

Question Number : 21 Question Id : 5623032981 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which feature of the calorimeter is designed to minimize losses due to radiation ?

Options :

1. The use of crucible arranged between two electrodes.
2. The construction of bomb with two valves.

3. The fitting of the calorimeter with a water jacket and further provision of an additional water and air jacket.

4. The use of thermometer to measure the temperature up to accuracy of 0.001°C .

**Question Number : 22 Question Id : 5623032982 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5**

What is the primary cause of slagging in furnace ?

Options :

1. Low furnace temperatures causing solidified fly ash deposits.

2. Accumulation of molten, partially fused or resolidified deposits on furnace walls exposed to radiant heat.

3. Formation of high temperature bonded deposits on convection heat-absorbing surfaces.

4. Clogging of gas passages with dry ash particles.

**Question Number : 23 Question Id : 5623032983 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5**

Which method is commonly used to measure the volumetric composition of combustion products such as CO_2 , CO and O_2 in flue gas ?

Options :

1. Haldane appartus
2. Infra-red gas analyzes
3. Gas chromatograph
4. Orsat apparatus

Question Number : 24 Question Id : 5623032984 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

_____ is described as a function of the ratio of turbulent burning velocity to the velocity of the fresh mixture entering the combustion zone.

Options :

1. Fuel performance
2. Boiler efficiency
3. Combustion performance

4. None of the above

Question Number : 25 Question Id : 5623032985 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The mass of excess air supplied is equal to _____.

Options :

1. $\frac{23}{100} \times \text{mass of excess carbon}$

2. $\frac{23}{100} \times \text{mass of excess oxygen}$

3. $\frac{100}{23} \times \text{mass of excess carbon}$

4. $\frac{100}{23} \times \text{mass of excess oxygen}$

Question Number : 26 Question Id : 5623032986 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which is not true for Supercritical boilers ?

Options :

1. The evaporator circuit is once through type.

2. The once through mode is operational at any load.
3. Super critical boiler operates in circulation mode.
4. None of the above.

Question Number : 27 Question Id : 5623032987 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

The swell phenomenon in the boiler drum leads to _____.

Options :

1. Increase in the feed water flow.
2. Decrease in the feed water flow.
3. No change in the feed water flow.
4. None of the above.

Question Number : 28 Question Id : 5623032988 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

The mechanical steam trap operates on which of the following principles ?

Options :

1. By sensing the difference in temperature between steam and condensate.
2. By sensing the difference in density between steam and condensate.
3. By sensing the difference in pressure between steam and condensate.
4. All of the above.

Question Number : 29 Question Id : 5623032989 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Why is it crucial to separate moisture from steam before it enters the superheater tubes ?

Options :

1. Moisture reduces steam pressure.
2. Moisture deposits salts that damage tubes.
3. Moisture lowers the superheating temperature.
4. Moisture increases the cost of superheater material.

Question Number : 30 Question Id : 5623032990 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In a natural circulation water tube boiler, from where does the water receive most of its heat for boiling ?

Options :

1. In the preheater where it receives heat from outgoing flue gases.
2. In the downcomer tubes carrying water from the steam drum.
3. In the water lining the furnace walls.
4. In the steam drum where it mixes with hot steam.

Question Number : 31 Question Id : 5623032991 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What is the primary purpose of an economiser in a steam generation system ?

Options :

1. To preheat boiler feedwater using steam from the high pressure turbines.
2. To recover heat from the flue gases to raise the feedwater temperature.
3. To separate moisture from steam before entering superheater.
4. To control the pressure of steam leaving the boiler drum.

Question Number : 32 Question Id : 5623032992 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What is the main benefit of using superheaters in a steam turbine cycle ?

Options :

1. They increase the moisture content in the last stages of the turbine.
2. They separate impurities from the steam before entering the turbine.
3. They improve overall efficiency by increasing steam temperature.
4. They control the pressure of the steam entering the turbine.

Question Number : 33 Question Id : 5623032993 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In a closed loop control system for a steam generator, what is the primary function of the error signal ?

Options :

1. To directly control the fuel firing rate.
2. To indicate the electrical power output of the plant.

3. To adjust the control system output based on the difference between desired and actual conditions.
4. To measure the steam flow rate leaving the boiler.

Question Number : 34 Question Id : 5623032994 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

What is the primary function of steam pressure control system, also known as “Boiler Master” ?

Options :

1. To maintain water level in the boiler.
2. To regulate steam temperature.
3. To maintain steam pressure by adjusting fuel and air flows.
4. To control the exhaust gas emissions.

Question Number : 35 Question Id : 5623032995 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

Which of the following methods for superheated steam temperature control in boiler offers the most efficient operation ?

Options :

1. Increasing excess combustion air.
2. Tilting burners upwards at low loads.
3. Using only upper burners at low loads.
4. Decreasing feed water temperature.

Question Number : 36 Question Id : 5623032996 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What is the primary benefits of using an air preheater in a steam generation system ?

Options :

1. To prevent acid corrosion in the flue gas exhaust system.
2. To improve the efficiency of boiler by recovering heat from the flue gas.
3. To maintain specific temperature for proper dispersion of exhaust gases.
4. To dry pulverized coal before combustion.

Question Number : 37 Question Id : 5623032997 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In cyclone separator, what principle is used to separate particles from a gas stream ?

Options :

1. Gravitational setting of particles.
2. Centrifugal forces generated by a spinning gas stream.
3. Filtration through porous membrane.
4. Electrostatic attraction between particles and electrodes.

Question Number : 38 Question Id : 5623032998 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What is the key difference in the general arrangement of a Supercritical boiler compared to a Conventional boiler ?

Options :

1. Use multiple drums.
2. Disposition of water and steam circuits.
3. Use of fire tubes.

4. Presence of an economiser.

Question Number : 39 Question Id : 5623032999 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In an Electrostatic Precipitator (ESP), how do dust particles in the gas stream acquire a charge ?

Options :

1. By direct contact with grounded collection electrodes.
2. By colliding with negatively charged gas molecules (ions).
3. By being exposed to the high voltage on discharged electrodes.
4. By absorbing light from the corona around the discharged electrodes.

Question Number : 40 Question Id : 5623033000 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What is the primary function of the degasifier in demineralization plant for boiler feed water treatment ?

Options :

1. To remove dissolved minerals that ion exchange resins cannot capture.

2. To remove carbon dioxide gas, which can be corrosive in the boiler system.
3. To regenerate the cation and anion exchange resins used in the demineralization process.
4. To remove silica from the water, which can damage turbine blades at high pressure.

Question Number : 41 Question Id : 5623033001 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Different fittings and devices necessary for operation and safety of the boiler are known as _____.

Options :

1. Mountings
2. Accessories
3. Both (1) and (2)
4. None of the above

Question Number : 42 Question Id : 5623033002 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which material is generally used for manufacturing lever safety valve ?

Options :

1. Bronze
2. Steel
3. Cast Iron
4. Brass

Question Number : 43 Question Id : 5623033003 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In Boilers of Lancashire and Cornish type, which of the following boiler mounting should be fitted ?

Options :

1. High pressure and low water safety valve.
2. Super heater.
3. Fusible plug.
4. Economiser.

Question Number : 44 Question Id : 5623033004 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which of the following statements accurately compares the function of injectors and economiser in a boiler system ?

Options :

1. Injectors use steam pressure to feed water and recover heat, while economiser preheat water using flue gas but don't recover heat.
2. Injectors are more efficient for larger boilers due to their simplicity, while economisers are complex and better suited for smaller boilers.
3. Injectors are steam powered feed water pumps, while economisers preheat feed water using waste heat flow flue gas.
4. Both injectors and economisers use external power source (electricity or motors) to function.

Question Number : 45 Question Id : 5623033005 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which of the following statement is true and valid for spring loaded safety valve ?

Options :

1. The control of the valve is not at all possible as it is dependent on steam pressure.
2. The control of the valve is held by the driver and the driver can release the pressure from either valve.
3. The spring loaded valve operates automatically depending upon spring pressure only.
4. The spring loaded safety valve operates automatically depending upon water pressure.

Question Number : 46 Question Id : 5623033006 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

The fitting mounted on the boiler, whose function is to put off the fire in the furnace, when level of water falls of an unsafe limit is called _____.

Options :

1. Safety valve
2. Stop valve
3. Fusible plug

4. Blow off cock

Question Number : 47 Question Id : 5623033007 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The function of feed check valve in boiler is _____.

Options :

1. To allow the supply of water to the boiler at high pressure continuously.
2. To prevent the back flow of water from boiler when pump pressure is low.
3. Both (1) and (2).
4. None of (1) and (2).

Question Number : 48 Question Id : 5623033008 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

For maximum discharge through Chimney which of the following is true ?

Options :

1. The height of the hot gas column is equal to the Chimney height.
2. The hot gas column is half that of Chimney height.

3. The hot gas column is twice that of Chimney height.
4. None of the above.

Question Number : 49 Question Id : 5623033009 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

The condition for maximum discharge through a Chimney is given by _____.

Options :

1. $(h_w)_{\max} = H \left(\frac{m_a}{m_a + 1} \cdot \frac{1}{T_a} \right) \text{ mm of water}$

2. $(h_w)_{\max} = H \left(\frac{m_a}{m_a + 1} \right) \text{ mm of water}$

3. $(h_w)_{\max} = \frac{353.5H}{T_a} \text{ mm of water}$

4. $(h_w)_{\max} = \frac{176.5H}{T_a} \text{ mm of water}$

Question Number : 50 Question Id : 5623033010 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

What is the height of Chimney required to produce a draught equivalent to 1.7 cm of water if the flue gas temperature is 270°C and ambient temperature is 22°C and minimum amount of air per kg of fuel is 17kg ?

Options :

1. 33.46 m
2. 27.29 m
3. 50.80 m
4. 57.34 m

Question Number : 51 Question Id : 5623033011 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In which of the following device the waste heat of the flue gases is utilised for heating the feed water ?

Options :

1. Preheater
2. Economiser
3. Injector

4. Superheater

Question Number : 52 Question Id : 5623033012 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

_____ is fitted to the crown plate of the furnace.

Options :

1. Economiser
2. Fusible plug
3. Superheater
4. Injector

Question Number : 53 Question Id : 5623033013 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

The air pressure at the fuel bed is reduced below that of atmosphere by means of a fan placed at or near the bottom of the Chimney to produce a draught. Such draught is called as which of the following ?

Options :

1. Natural draught

2. Induced draught
3. Forced draught
4. Balanced draught

Question Number : 54 Question Id : 5623033014 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which valve is used to remove sludge and sediment from the bottom of a boiler ?

Options :

1. Safety valve
2. Blowdown valve
3. Globe valve
4. Gate valve

Question Number : 55 Question Id : 5623033015 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The velocity of flue gases (V) through the Chimney under a static draught of H' meters is given by _____.

Options :

1. $4.43H'$
2. $(4.43H')^2$
3. $4.43(H')^2$
4. $4.43 \sqrt{H'}$

Question Number : 56 Question Id : 5623033016 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

Starting of a boiler can be classified in _____.

Options :

1. three types
2. two types
3. four types
4. five types

Question Number : 57 Question Id : 5623033017 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

Which of the following, is not a part of forced shutdown of a boiler ?

Options :

1. drum level is too high.
2. oil-soot burns in the air heater.
3. there is violent vibration, which can't be stopped.
4. if steam or feed lines are working properly.

Question Number : 58 Question Id : 5623033018 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Preventive maintenance takes time to show up the results but _____ is immediately effective.

Options :

1. breakdown maintenance
2. scheduled maintenance
3. productive maintenance
4. maintenance prevention

Question Number : 59 Question Id : 5623033019 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In most demanding Emergency Shut Down (ESD) applications of high temperature and high differential pressure _____ is recommended.

Options :

1. Fully metal-sealed trunnion mounted ball valve
2. Butterfly valve
3. Dead-weight safety valve
4. None of the above

Question Number : 60 Question Id : 5623033020 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

When the enthalpy or total heat of steam is h KJ/Kg and enthalpy or sensible heat of feed water is hf_1 KJ/Kg, then the factor of evaporation is given by _____.

Options :

1. $\frac{h + hf_1}{2257}$

2. $\frac{h}{hf_1 \times 2257}$

3. $\frac{h \times hf_1}{2257}$

4. $\frac{h - hf_1}{2257}$

Question Number : 61 Question Id : 5623033021 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

What is the purpose of record keeping in boiler maintenance ?

Options :

1. To monitor fuel consumption
2. To ensure compliance with safety regulations
3. To track maintenance activities and identify trends or recurring issues
4. All of the above

Question Number : 62 Question Id : 5623033022 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

A typical value of boiler turn down is _____.

Options :

1. 0.4 : 1
2. 40 : 1
3. 4 : 1
4. None of the above

Question Number : 63 Question Id : 5623033023 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The highest heat loss in the boiler is which among the following ?

Options :

1. Heat loss due to evaporation of water formed due to H_2 in fuel.
2. Heat loss due to moisture in fuel.
3. Heat loss due to dry flue gases.
4. None of the above.

Question Number : 64 Question Id : 5623033024 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

How many fundamental control strategies are used in process control of a boiler ?

Options :

1. 3
2. 5
3. 7
4. 9

Question Number : 65 Question Id : 5623033025 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Loss of heat to evaporate the moisture (H8) = _____, in boiler heat balance calculations.

If M_m = mass of moisture formed by burning of hydrogen per kg of fuel,
kg H_2O /kg fuel.

L = Latent heat of evaporation of moisture at the dew point of flue gases Kcal/Kg.

Options :

1. $\frac{M_m}{L}$

2. $M_m \times L$

3. $\frac{M_m \times L}{1 + M_m}$

4. None of the above

Question Number : 66 Question Id : 5623033026 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The ratio of the heat absorbed by the BFW in the economizer to the heat supplied by the flue gases in the economizer, the temperature of the flue gases being reckoned above the temperature of air supplied to the boiler, is called _____.

Options :

1. boiler efficiency

2. equivalent evaporation

3. economizer efficiency

4. none of the above

Question Number : 67 Question Id : 5623033027 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Determine the boiler efficiency from the given data :

Heat utilized = 6000 Kcal/Kg Coal

Excess air supplied = 40%

Flue gas temperature at boiler exit = 170°C

Heat generated = 8000 Kcal/Kg Coal.

Options :

1. 78%

2. 73%

3. 75%

4. None of the above

Question Number : 68 Question Id : 5623033028 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In practice, a large number of items that affect the boiler efficiency are related to _____.

Options :

1. excess water feed
2. excess heat supply
3. excess air
4. excess fuel supply

Question Number : 69 Question Id : 5623033029 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The requirement of excess air in a boiler varies depending upon _____.

Options :

1. the fuel
2. the burner design
3. the furnace design
4. all of the above

Question Number : 70 Question Id : 5623033030 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which device is used to reduce the temperature of superheated steam ?

Options :

1. Air preheater
2. Economizer
3. De-superheater
4. Soot blower

Question Number : 71 Question Id : 5623033031 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

As per Reg. 148, of the Indian Boiler Regulations, 1950, minimum thickness required for 70 mm on stay tube fitted in smoke tube boilers is _____.

- a. 3.25 mm
- b. 3.40 mm
- c. 3.12 mm
- d. 3.66 mm

Options :

1. c only
2. a only
3. b only

4. d only

Question Number : 72 Question Id : 5623033032 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Maximum feed water temperature at outlet of the economiser shall be _____.

Options :

1. equal to the saturated steam temperature in boiler drum.
2. atleast 40°C below the saturated steam temperature in boiler drum.
3. atleast 40°F below the saturated steam temperature in boiler drum.
4. 40°F above the saturated steam temperature in boiler drum.

Question Number : 73 Question Id : 5623033033 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

As per the Boilers Act, a certificate authorising the use of boiler shall cease to be in force _____.

- a. When an accident occurs to the boiler.
- b. When any structural alteration or renewal is made to the boiler with prior permission of the Chief Inspector of Boiler.
- c. On expiry of certificate for which it was granted.
- d. When boiler is moved to other place.

Which of the statements given above is/are correct ?

Options :

- 1. a only
- 2. b and c only
- 3. a, c and d only
- 4. c only

Question Number : 74 Question Id : 5623033034 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

As per Section 29 of the Boilers Act, 1923, the State Government may make rules consistent with the Boilers Act and Regulations made thereunder, for following purposes, _____.

- a. The powers and duties of the Chief Inspector, Deputy Chief Inspectors and Inspector.
- b. The qualification and experience of persons to be appointed as Chief Inspector, Deputy Chief Inspector and Inspector.
- c. Fee payable for registration.
- d. For requiring boilers to be under the charge of persons holding certificate of proficiency or competency.

Options :

- 1. a and b
- 2. b and d
- 3. c and d
- 4. a and c

As per the Boilers Act, 1923 “Steam Pipe” means any pipe through which steam passes, if _____.

- a. The pressure at which steam passes through such pipe exceeds 1.0 kilogram per square centimeters above atmospheric pressure.
- b. Such pipe exceeds 254 millimeters in internal diameter and the pressure of steam exceeds 1 kilogram per square centimeters above the atmospheric pressure.
- c. The pressure at which steam passes through such pipe exceeds 3.5 kilogram per square centimeters above atmospheric pressure.
- d. Such pipe exceeds 254 millimeters in outer diameter and the pressure of steam exceeds 1 kilogram per square centimeters above the atmospheric pressure.

Which of the statement given above is /are correct ?

Options :

- 1. a only
- 2. a and c only
- 3. b and c only

4. c and d only

Question Number : 76 Question Id : 5623033036 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

As per the Boilers Act, 1923, “Competent Authority” means an institution recognised in such manner as may be prescribed by regulations for issue of _____.

Options :

1. Certificate to use the boiler.
2. Certificate of registration of boiler.
3. Certificate to the welders for welding of boilers and boiler components.
4. Certificate to boiler operation engineers and boiler attendants.

Question Number : 77 Question Id : 5623033037 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

According to Boilers Act, which of the following is not a Boiler ?

- a. Lancashire Boiler.
- b. A pressure vessel less than 25 liters in capacity.
- c. Fluidised bed boiler.
- d. A pressure vessel in which water is heated below 100°C.

Options :

- 1. c only
- 2. b and d only
- 3. a only
- 4. none of the above

**Question Number : 78 Question Id : 5623033038 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5**

Sterilizer disinfector used in the hospitals will fall under the provisions of the Boilers Act, 1923 if its capacity _____.

- a. exceeds 22.5 liters
- b. exceeds 25 liters
- c. exceeds 91 liters
- d. exceeds 100 liters

Options :

- 1. d only
- 2. b only
- 3. c only
- 4. a only

Question Number : 79 Question Id : 5623033039 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What does Section V of the ASME Boiler and pressure vessel code address ?

Options :

- 1. Non destructive examination

2. Welding and brazing qualifications
3. Material specifications
4. Construction of pressure vessels

Question Number : 80 Question Id : 5623033040 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

Among others, Central Boilers Board consists of representatives of _____.

- a. Bureau of Indian standards.
- b. Boiler and Boiler component manufacturers.
- c. Representatives of users of boilers.
- d. National Laboratories Ltd.

Options :

1. a only
2. c only
3. a and c only
4. All of the above

Question Number : 81 Question Id : 5623033041 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The cause of superheater tube failure due to accumulation of deposits in it : _____.

Options :

1. Carry over.
2. Excessive height of water maintained in the drum.
3. Deposits accumulated on steam separators in the drum.
4. All of the above.

Question Number : 82 Question Id : 5623033042 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What is the main cause of waterside corrosion in boilers ?

Options :

1. Dissolved oxygen
2. Sulphur dioxide
3. Carbon dioxide

4. Humidity

Question Number : 83 Question Id : 5623033043 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In boiler inspection, the hammer test is used _____.

- a. to detect wasted and cracked area.
- b. to determine the tightness of rivets, bolts and brackets.
- c. to discover the reduced wall thickness of boiler components.
- d. to determine the tube material.

Which of the statements given above is/are correct ?

Options :

- 1. a only
- 2. a and b only
- 3. a, b and c only
- 4. All of the above

Question Number : 84 Question Id : 5623033044 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What is the cause of boiler tube leakages due to creep rupture ?

Options :

1. Poor weld quality and joint integrity
2. Pitting and corrosion of tubes due to dissolved gases
3. High thermal stresses due to long term overheating
4. Mechanical fatigue due to repeated loading cycles

Question Number : 85 Question Id : 5623033045 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In a Boiler with water wall furnace maximum heat loss happens due to _____.

- a. Unburnt fuel
- b. Radiation
- c. Heat in flue gas leaving the furnace
- d. Continuous blow down

Options :

1. a and d only

2. b only
3. c only
4. d only

Question Number : 86 Question Id : 5623033046 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

In a boiler heat transfer takes place in which of the following methods ?

Options :

1. Radiation
2. Convection
3. Conduction
4. All of the above

Question Number : 87 Question Id : 5623033047 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

As per Indian Boiler Regulation 376(fff) appendix – JB, waste heat Boilers upto age of 20 years are inspected as per regulation-390. After satisfactory Inspection, Boiler are certified for a period of _____.

Options :

1. Twelve months
2. Eighteen months
3. Twenty four months
4. None of the above

Question Number : 88 Question Id : 5623033048 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

During inspection at the manufacturer's premises, fusion welded drum shall be rejected in the following cases; _____.

- a. Deviation from circularity in any section exceeds 2% in case of stress relieved drums.
- b. Straight line deviation exceeds 0.5% of the cylindrical length.
- c. Deviation from circularity in any section exceeds 2% incase of normalized drums.
- d. Deviation from circularity in any section exceeds 1% in case of stress relieved drums.

Options :

1. only a and b

2. only a and c
3. only b and c
4. only c and d

Question Number : 89 Question Id : 5623033049 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

During repairs of super heater coils by process of SMAW, for T 22 grade, following weld - consumables shall be used _____.

- a. Electrode E 8018 – B2
- b. Electrode E 9015 – B3
- c. Electrode E 9018 – B3
- d. Electrode E 8016 – B2

Options :

1. Either a or d
2. Either b or c
3. Either b or d
4. All of the above

Question Number : 90 Question Id : 5623033050 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

Which parameter in a tensile test represents the stress at which deformation changes from elastic to plastic behaviour ?

Options :

1. Ultimate Tensile Strength (UTS)
2. Young's modulus (E)
3. Yield Point (YP)
4. Modulus of Resilience (MR)

Question Number : 91 Question Id : 5623033051 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

At elevated temperatures even with constant load, material elongates over time. This property is known as _____, which is proportional to the load and the temperature.

Options :

1. Toughness
2. Hardness

3. Creep

4. Compressive strength

Question Number : 92 Question Id : 5623033052 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

The special heat treatment given to medium carbon, high carbon and low alloy steel wire rod is called _____.

Options :

1. Sub-zero treatment

2. Austempering

3. Martempering

4. Patenting

Question Number : 93 Question Id : 5623033053 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

_____ is primarily a very high-quality, fully deoxidised, mild steel plate of medium to high thickness, suitably alloyed where necessary to withstand high pressure and temperature.

Options :

1. ASME steel
2. Boiler quality plate
3. Copper
4. None of the above

Question Number : 94 Question Id : 5623033054 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

What is creep in its simplest form ?

Options :

1. Sudden failure under stress
2. Progressive accumulation of plastic strain under stress at elevated temperature
3. Immediate elastic deformation under stress
4. Reversible deformation under stress

Question Number : 95 Question Id : 5623033055 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

What are the types of cylindrical shells used in fabricating pressure vessels for the boilers ?

Options :

1. Monolithic cylindrical shells, multi layered shrink fit concentric cylindrical shells, filament wound cylindrical shells, wire wrapped cylindrical shells.
2. Single layer cylindrical shells, polymer wrapped cylindrical shells, fiber reinforced cylindrical shells, mesh wrapped cylindrical shells.
3. Metal cylindrical shells, plastic cylindrical shells, ceramic cylindrical shells, composite cylindrical shells.
4. Thick walled cylindrical shells, thin walled cylindrical shells, semi-ellipsoidal cylindrical shells, hemispherical cylindrical shells.

Question Number : 96 Question Id : 5623033056 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

_____ is the measure of the depth of full hardness achieved during heat treatment and it is not to be confused with hardness.

Options :

1. Hardenability
2. Wear resistance
3. Shear strength
4. Tensile strength

Question Number : 97 Question Id : 5623033057 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

_____ is a special welding technique which is basically used for butt welding of steel tubes at a temperature below the melting point of parent metal.

Options :

1. fore-hand welding
2. back-hand welding
3. linde welding
4. none of the above

Question Number : 98 Question Id : 5623033058 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

In heat treatment, austempering process is also known as _____.

Options :

1. stepped quenching process.
2. interrupted quenching process.
3. isothermal quenching process.
4. tempering process.

Question Number : 99 Question Id : 5623033059 Question Type : MCQ Option Shuffling : No
Correct Marks : 2 Wrong Marks : 0.5

What factors, in addition to pressure must be considered when analyzing the structural components of a boiler ?

Options :

1. Temperature, cycling, corrosion, creep and material properties
2. Pressure, vacuum, load and stress concentration

3. Volume, mass, density and specific heat
4. Flow rate, velocity, acceleration and displacement

Question Number : 100 Question Id : 5623033060 Question Type : MCQ Option Shuffling : No

Correct Marks : 2 Wrong Marks : 0.5

_____ is the most difficult of all gas welding positions and becoming proficient will take a great deal of practice.

Options :

1. Vertical welding
2. Horizontal welding
3. Inclined welding
4. None of the above